

Simple Yet Effective Way for Improving the Performance of Lossy Image Compression

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Abstract—Lossy image compression methods with deep neural network (DNN) include a quantization process between encoder and decoder networks as an essential part to increase the compression rate. However, the quantization operation impedes the flow of gradient and often disturbs the optimal learning of the encoder, which results in distortion in the reconstructed images. To alleviate this problem, this paper presents a simple yet effective way that enhances the performance of lossy image compression without imposing training overhead or modifying the original network architectures. In the proposed method, we utilize an auxiliary branch called a shortcut which directly connects the encoder and decoder. Since the shortcut does not include the quantization process, it supports the optimal learning of the encoder by flowing the accurate gradient. Furthermore, to assist the decoder which should handle additional feature maps obtained via the shortcut, we also propose a residual refinement unit (RRU) following the quantizer. The experimental results show that the image compression network trained with the proposed method remarkably improves the performance in terms of peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and multi-scale structural similarity (MS-SSIM).

Index Terms—Convolutional neural network, deep learning, image compression.

I. INTRODUCTION

DEEP neural network (DNN)-based lossy image compression methods [1]–[9] have been widely developed. As an early work of these techniques, Toderici *et al.* [1], [2] proposed novel network architectures based on a convolutional recurrent neural network (RNN). Although these methods outperform the traditional image coding standards such as JPEG [10] and JPEG2000 [11], they require considerable processing time since the image is encoded and decoded in an iterative manner.

For fast image compression, recently, various methods employing a convolutional neural network (CNN) have been proposed [3]–[7]. These methods generally consist of the encoder, quantizer, and decoder. In particular, the quantizer is an

essential component to compress the feature with a low bit-rate. However, the encoder cannot be trained since it is located before the quantizer whose operation is non-differentiable. To solve this problem, Theis *et al.* [3] utilized a smooth approximation in the backward pass for dealing with the non-differentiability of the rounding operation, whereas Ballé *et al.* [4] replaced the quantization error with uniform noise during the training procedure. To further enhance the compression performance, Li *et al.* [6] proposed a content-weighted compression network which determines the pixel-wise different bit allocation by considering the edge or texture of the original image. For very low bit-rate coding, Agustsson *et al.* [9] introduced a generative adversarial network (GAN)-based compression model which applies the segmentation map to synthesize non-important regions. In addition, to compress the feature holding the information of the image is widely employed in many vision tasks [12]–[15]. The aforementioned CNN-based methods [3]–[8] commonly employ differentiable approximations for backward propagation. However, these approximations could cause the unsuitable calculation of the gradient since the approximation results often differ from the quantized values.

This paper presents a simple yet effective joint learning method which significantly enhances the performance of image compression. The proposed method employs a two-branch network consisting of the main branch with the quantizer and an auxiliary branch, called shortcut, without the quantizer. The encoder can be accurately trained by back-propagating the appropriate gradient via the shortcut since the shortcut is not involved in the quantization process. However, the two-branch network forces the decoder to handle both quantized feature maps with binary numbers and non-quantized ones with real numbers. To prevent the side-effect of the shortcut, we also propose a residual refinement unit (RRU) that converts the quantized feature maps into non-quantized-like feature maps. Since the proposed RRU requires a small number of network parameters, it can be readily applied to existing frameworks with marginal training overheads and modifications. Through extensive experiments on several datasets including the AVA database [16], the Berkeley segmentation dataset (BSD) [17], and the LIVE dataset [18], the performance of the proposed method is compared with that of an existing method in terms of image quality assessment (IQA) metrics: peak signal-to-noise ratio (PSNR), structural similarity (SSIM) [19], and multi-scale structural similarity (MS-SSIM) [20].

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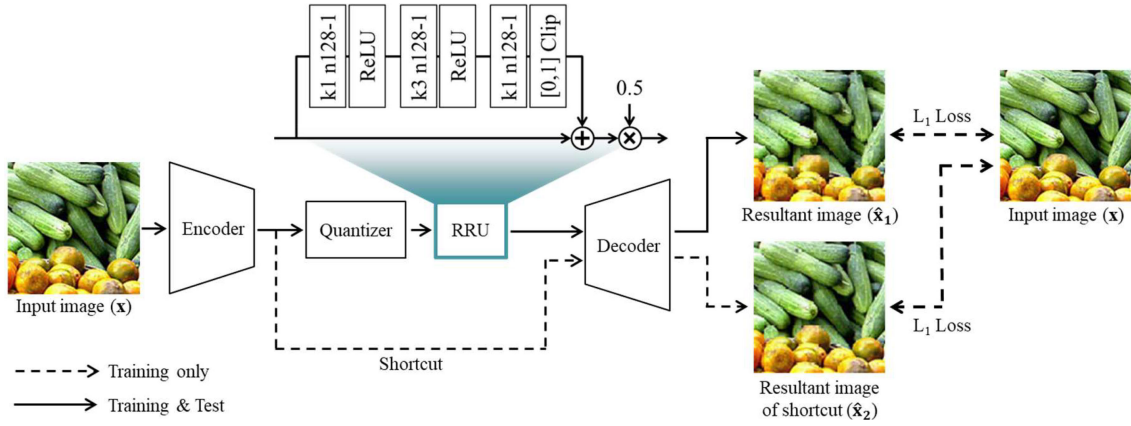


Fig. 1. Architecture of the proposed joint learning method. Through two-branch including the shortcut, the whole network is trained only with the L_1 loss.

II. PROPOSED METHOD

A. Two-Branch Network

Fig. 1 illustrates the framework of the proposed joint learning method. Given an input image, the encoder first extracts the feature maps using multiple convolutional layers. For low bit-rate image coding, the encoded feature maps with real numbers require the quantization process. However, as mentioned in the previous section, the quantization process often interferes with the optimal learning of the encoder. Indeed, in the forward propagation, the encoder aims to transform the input image $\mathbf{x}$ into the non-quantized feature maps $\mathbf{f}$ which are suitable for minimizing the objective function O , such as L_1 loss. In other words, to achieve the optimal results, the encoder should be trained with the accurate gradient in the backward propagation. However, the encoder employs the improper gradient caused by the non-differentiable quantization operation. The improper gradient is formulated as

$$\frac{\partial O}{\partial \mathbf{x}} = \frac{\partial O}{\partial \mathbf{f}} \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \frac{\partial O}{\partial \mathbf{q}} \frac{\partial \mathbf{q}}{\partial \mathbf{f}} \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = k \cdot \frac{\partial O}{\partial \mathbf{q}} \frac{\partial \mathbf{f}}{\partial \mathbf{x}}, \quad (1)$$

where q indicates the quantized feature maps. Since the $\partial \mathbf{q} / \partial \mathbf{f}$ is generally approximated to a constant value k , e.g., $k = 1$ in [6], the encoder cannot use the optimal gradient during the training procedure; the approximated gradient drives the inappropriate learning.

To alleviate this problem, we employ the shortcut which provides the output of the encoder to the decoder. It allows the decoder to directly handle the feature maps without quantization. During the training procedure, the shortcut helps the encoder to be accurately trained by conveying the optimal gradient. To train the two-branch network, we propose a loss function that maximizes the effectiveness of the shortcut. As depicted in Fig. 1, the proposed network produces two diff resultant images, i.e., $\hat{\mathbf{x}}_1$ and $\hat{\mathbf{x}}_2$, from the main and auxiliary branches, respectively. To jointly minimize the pixel-wise mean absolute error (MAE) between the input image and aforementioned $\hat{\mathbf{x}}_1$ and $\hat{\mathbf{x}}_2$, the loss function is defined as

$$L = \|\mathbf{x} - \hat{\mathbf{x}}_1\| + \lambda \|\mathbf{x} - \hat{\mathbf{x}}_2\|, \quad (2)$$

where λ plays the role of hyper-parameter that controls the contribution of the auxiliary branch. The hyper-parameter λ

stays at 1 (or 0) when the iteration number of the training process is less (or more) than half of its maximum. In the first half of the training procedure, the proposed loss function guides the encoder to be accurately trained via the shortcut by minimizing the distortion of $\hat{\mathbf{x}}_2$. In the second half of the training process, the proposed network focuses on minimizing the distortion of $\hat{\mathbf{x}}_1$ for achieving image compression with high quality.

B. Residual Refinement Unit (RRU)

On this occasion, the shortcut often places a little burden on the learning of the decoder since it should handle two different types of feature maps: the quantized and non-quantized feature maps which consist of binary and real numbers, respectively. To address this problem, we propose an RRU which converts the quantized feature maps into the non-quantized-like ones. Since the quantized feature maps are obtained by a rounding operation, we design the RRU to re-convert the values of the quantized feature maps, i.e., 0 and 1, into real numbers in the range of $[0, 0.5]$ and $[0.5, 1]$, respectively. In a detailed architecture of the RRU, “k1 n128-1” represents a convolution layer with kernel size 1, 128 output channels and a stride of 1. In the residual branch of the RRU, the quantized feature maps are processed through multiple convolutional and rectified linear unit (ReLU) layers, and then clipped into $[0, 1]$. In that branch, both 0 and 1 of the quantized feature maps are converted to real numbers in the range of $[0, 1]$. To convert 0 (or 1) into a real number less (or more) than 0.5, we average the input and output of the residual branch. The RRU consequently generates an output value whose range is adjusted according to the input value; this procedure only needs to modify the quantized feature maps without any specific constraints.

C. Implementation Details

In this paper, we trained the entire network for 400 k iterations using a batch size of 8 in an end-to-end manner. We employed Adam optimizer [21] which is a stochastic optimization method with adaptive estimation of moments. The hyper-parameters of the Adam optimizer β_1 and β_2 were set to 0.9 and 0.999, respectively. We set an initial learning rate to 0.0001 and reduced the learning rate to 1/10 after 300k iterations. Our experiments were conducted on CPU Intel(R) Xeon(R) CPU E3-1245 v5 and

TABLE I
EXPERIMENTAL RESULTS OBTAINED BY APPLYING THE PROPOSED METHOD TO THE BASELINE MODELS [3], [6], [8]

Data set	Baseline model	Shortcut + RRU	0.125 bpp		0.25 bpp		0.5 bpp		1.0 bpp		2.0 bpp	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
BSD [17]	Theis [3]	✓	23.365	0.641	24.767	0.716	26.466	0.796	28.160	0.858	29.481	0.894
			23.602	0.648	24.989	0.722	26.626	0.800	28.438	0.864	30.170	0.910
	Li [6]	✓	23.165	0.628	24.768	0.708	26.246	0.772	27.215	0.804	27.718	0.819
LIVE [18]			23.417	0.640	24.936	0.722	26.564	0.797	28.097	0.851	29.133	0.878
	Aytekin [8]	✓	22.995	0.620	24.444	0.699	26.159	0.787	28.006	0.854	29.509	0.895
			23.208	0.636	24.673	0.713	26.339	0.794	28.178	0.860	29.968	0.907
LIVE [18]	Theis [3]	✓	22.790	0.640	24.286	0.718	26.062	0.798	27.806	0.858	29.096	0.892
			23.102	0.647	24.540	0.725	26.273	0.802	28.108	0.864	29.785	0.907
	Li [6]	✓	22.665	0.626	24.292	0.708	25.833	0.773	26.780	0.806	27.269	0.820
LIVE [18]			22.924	0.637	24.548	0.724	26.198	0.799	27.748	0.852	28.793	0.879
	Aytekin [8]	✓	22.510	0.619	23.988	0.700	25.782	0.790	27.677	0.856	29.164	0.895
			22.644	0.635	24.080	0.714	25.973	0.796	27.856	0.861	29.621	0.905

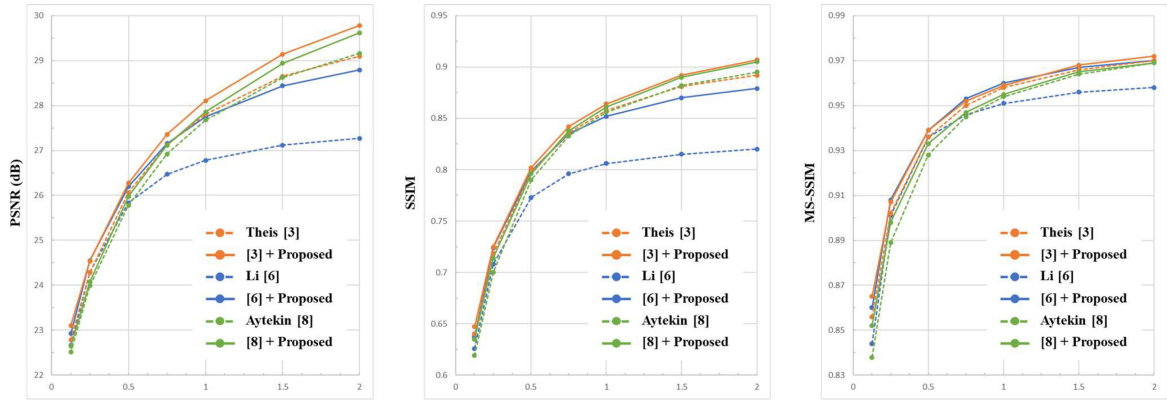


Fig. 2. Rate distortion curves on the LIVE dataset given as PSNR, SSIM, and MS-SSIM vs. bit per pixel (bpp).

GPU TITAN X (Pascal), and implemented in *TensorFlow* v1.8. For our experiments, we trained our network using the AVA database and tested it on the LIVE dataset and the BSD. For training of the network, we selected 50,000 high-quality images with high subjective scores from the AVA database and randomly sampled three 128×128 patches from each image. During the test procedure, we employed the original images from both the LIVE dataset and the BSD as test images.

III. EXPERIMENTAL RESULTS

In this section, we show qualitative and quantitative comparison results to demonstrate the superiority of the proposed method. To evaluate the effectiveness of the proposed method, we adopted the encoder and decoder architectures from the leading DNN-based image compression frameworks [3], [6], [8].

A. Quantitative Comparison

We evaluated the performance of the proposed method by measuring PSNR and SSIM. We conducted extensive experiments by adjusting the channels of the encoded feature maps from 8 to 128; these channels correspond to 0.125 to 2.0 bits per pixel (bpp).

TABLE II
PERFORMANCE ON THE LIVE DATASET MEASURED AS AREA UNDER THE CURVE (AUC), UP TO 2.0 BPP

Baseline model	Shortcut + RRU	PSNR	AUC SSIM	AUC MS-SSIM
Theis [3]		49.465	1.479	1.766
	✓	51.196	1.558	1.783
Li [6]		51.302	1.568	1.779
	✓	52.010	1.584	1.784
Aytekin [8]		51.062	1.561	1.769
	✓	51.524	1.576	1.774

To confirm the generalization ability of the proposed method, we measured the performance of the aforementioned three image compression baseline models with/without the proposed method composed of the shortcut and RRU. Results are shown in Table I, where ✓ indicates that the baseline model is trained with the proposed learning technique. The proposed method significantly improves the performance of all the baseline models. Fig. 2 and Table II show that all the models benefit from the proposed method in terms of PSNR, SSIM, and MS-SSIM. Additionally, Fig. 3 depicts the test losses over the training iterations. It shows

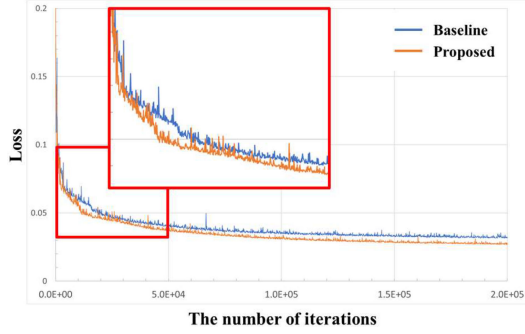


Fig. 3. A graph of the test losses on the LIVE dataset with respect to the number of iterations.

TABLE III
EXPERIMENTAL RESULTS OBTAINED BY APPLYING THE SHORTCUT AND/OR THE RRU TO THE BASELINE MODEL [6] ON PSNR

Data set	Short cut	RRU	bpp				
			0.125	0.25	0.5	1.0	2.0
BSD [17]	✓		23.165	24.768	26.246	27.215	27.718
			23.359	24.866	26.556	28.045	29.012
		✓	23.143	24.814	26.337	27.349	27.896
	✓	✓	23.417	24.936	26.564	28.097	29.133
LIVE [18]			22.665	24.292	25.833	26.780	27.269
		✓	22.849	24.419	26.169	27.683	28.664
		✓	22.676	24.387	25.920	26.930	27.470
	✓	✓	22.924	24.548	26.198	27.748	28.793

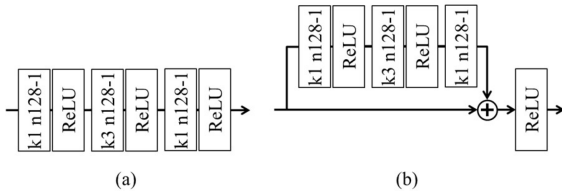


Fig. 4. The illustration of the fundamental blocks. (a) Block 1 (b) Block 2.

that the proposed method effectively drives the baseline model to optimal learning, especially in the early stages of the training procedure.

To verify the effectiveness of the shortcut and RRU, we also performed ablation studies removing either shortcut or RRU from the proposed method. As depicted in Table III, the proposed models trained without the shortcut mostly outperform the baseline while exhibiting considerably poor performance compared with the full models. These observations indicate that the shortcut encourages the encoder to be trained accurately. In addition, the proposed method without RRU shows a decrease in performance. These results demonstrate that the RRU helps the decoder handle the feature maps obtained via two-branch.

Furthermore, we conducted several extra-experiments with other network design choices. Instead of the RRU, we employed some fundamental blocks, i.e., block 1 and 2, having the same number of network parameters. As shown in Fig. 4, block 1 is a series of stacked convolutional layers and the other one is a general residual block adopted from [22]. As shown in Table IV, the proposed method with RRU achieves superior performance.

TABLE IV
COMPARISON WITH DIFFERENT BLOCKS ON THE LIVE DATASET

Method	[0, 1] clipping	1.0 bpp		2.0 bpp	
		PSNR	SSIM	PSNR	SSIM
RRU	✓	27.748	0.852	28.793	0.878
Block 1		27.251	0.831	27.892	0.849
	✓	27.386	0.833	28.009	0.850
Block 2		25.824	0.765	26.683	0.801
	✓	26.178	0.779	26.639	0.797

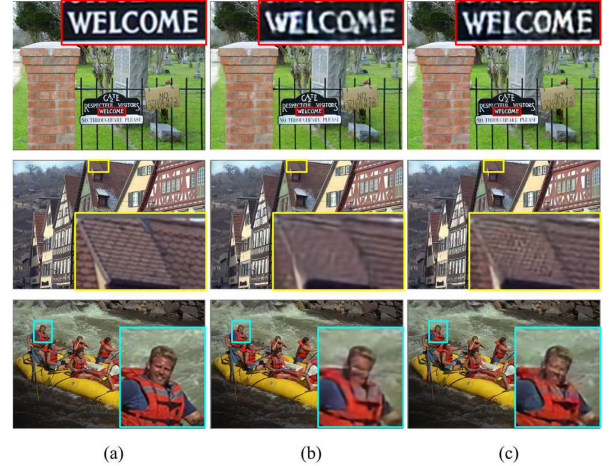


Fig. 5. Visual quality comparison on the LIVE dataset (1.0 bpp). (a) Original images. (b) Reconstructed images of the baseline model. (c) Reconstructed images of the proposed method.

It reveals that the RRU has a meaningful structure for the stable learning of the decoder.

B. Qualitative Comparison

We compared the visual quality of the reconstructed images obtained by the baseline model [6] and the proposed method. As depicted in Fig. 5, the baseline model usually shows artifacts such as blurring in the strong edges and over-smoothing the small-scale textures. For example, in the first row, the details of the letters are blurred in the baseline model. In the second row, most textures of the roof are removed. On the other hand, the proposed method well reconstructs the small-scale textures and details in the original images. These results demonstrate that the proposed method effectively reduces the distortion in the reconstructed images.

IV. CONCLUSION

In this paper, we introduced a novel joint learning method for DNN-based image compression. Using the joint learning method with both shortcut and RRU, we effectively improve the performance of the existing image compression frameworks in terms of PSNR, SSIM, and MS-SSIM. It is expected that the proposed method is capable to improve the performance of the various DNN-based image compression methods without imposing training overhead or modifying the original network architectures. As our future work, we will adopt the proposed joint learning method to other low-level vision problems such as low-light image enhancement and image denoising.

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